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Safety and Efficacy of Recombinant Botulinum Toxin Type A for Moderate-to-severe Glabellar Lines: A Multicenter Randomized Controlled Trial

Running Title: Recombinant BoNT/A for Glabellar Lines

Qingfeng Li, MD[1], Xiao Long, MD[2], Chongyuan Xu, MD[3], Zhiqi Hu, MD[3], Yun Xie, MD[1], Nanze Yu, MD[2], Ping Jiang, MD[3], Wu Yang, PhD[4], Xiaoyun Sun, MD[4*]

[1]Shanghai Ninth People's Hospital, Shanghai Jiao Tong University School of Medicine, 639 Zhizaoju Road, Shanghai, China;

[2]Peking Union Medical College Hospital (Dongdan campus), No.1 Shuaifuyuan Wangfujing Dongcheng District, Beijing, China;

[3]Nanfang Hospital, Southern Medical University, No.1838, North Guangzhou Avenue, Guangzhou, China;

[4]Chongqing Claruvis Pharmaceutical Co., Ltd., No.28, Yuefu Avenue, DaDi (International) Life Science Park, A2, Beibei District, Chongqing, China.

***Corresponding author: Xiaoyun Sun, MD.** Chongqing Claruvis Pharmaceutical Co., Ltd., No.28, Yuefu Avenue, DaDi (International) Life Science Park, A2, Beibei District, Chongqing, China. Tel: +86-13810399273. E-mail: magnolia_sun@163.com. E-mail: xiaoyun.sun@claruvis.com.

Disclosure Statement

Funding: This study was sponsored by Chongqing Claruvis Pharmaceutical Co., Ltd.

Conflict of Interest: Wu Yang and Xiaoyun Sun reported being an employee and shareholder of Chongqing Claruvis. Other authors have no conflict of interest.

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Declarations

Ethical Statement: The study was conducted in accordance with the ethical principles of the Declaration of Helsinki, in compliance with Chinese Good Clinical Practice and the International Conference on Harmonisation-Good Clinical Practice (ICH-GCP), and was initially approved by the Biomedical Research Ethics Committee of Shanghai Ninth People's Hospital, Shanghai JiaoTong university School of Medicine (SH9H-2022-C18-3), followed by subsequently approval from the Ethics Committee of other study sites. All participants signed written informed consent before screening.

Author contributions: Qingfeng Li: Conceptualization, Methodology, Data curation, Formal analysis, Project administration, Resources, Writing – original draft, Writing – review & editing; **Xiao Long:** Data curation, Formal analysis, Writing – review & editing; **Chongyuan Xu**: Data curation, Formal analysis, Writing – review & editing; **Zhiqi Hu:** Data curation, Formal analysis, Writing – review & editing; **Yun Xie:** Data curation, Formal analysis, Writing – review & editing; **Nanze Yu:** Data curation, Formal analysis, Writing – review & editing; **Ping Jiang:** Data curation, Formal analysis, Writing – review & editing; **Wu Yang:** Conceptualization, Methodology, Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing; **Xiaoyun Sun:** Conceptualization, Methodology, Data curation, Formal analysis, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing.

Data availability statement: The datasets used and/or analyzed in the current study are available from the corresponding author upon reasonable request.

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Abstract

Background: YY001 (Retoxin®), the first global recombinant botulinum toxin type A (rBoNT/A) tested in humans, contains a pure 150-kDa core neurotoxin identical to natural BoNT/A (nBoNT/A). This study aimed to evaluate the safety and preliminary efficacy of YY001 in humans for moderate-to-severe glabellar lines.

Methods: Preclinical assessments comprised homology modeling, biochemical and pharmacological evaluation, and multispecies toxicology. In this phase I multicenter double-blind, randomized, dose-ranging, vehicle- and active- controlled trial, participants (n=50) with moderate-to-severe glabellar lines were randomized 1:1:1:1:1 to receive intramuscular injection of YY001 at 10, 20 and 30 U, Botox® at 20 U, or vehicle. The primary endpoint encompassed the incidence of adverse events (AEs), serious AEs, injection site reactions and clinically significant laboratory changes within 28 days post-injection, with exploratory follow-up to 168 days.

Results: The YY001's structure and biological activities are highly consistent with those of nBoNT/A. In this trial, no drug-related AEs were observed in the tested 10-30 U of YY001. At day 28, the response rate by investigator's live assessment was 60% for 10 U of YY001, 80% for 20 U of YY001, 90% for 30 U of YY001, 70% for 20 U of Botox®, and 0% for the vehicle, while the proportion of subjects with none or mild glabellar lines at maximum frown was 80%, 90%, 100%, 80%, and 0%, respectively.

Conclusions: The first rBoNT/A (YY001) in human study demonstrated a favorable safety and tolerability profile (dose-independent) at doses of 10, 20 and 30 U in Chinese subjects with moderate-to-severe glabellar lines. It also showed dose-dependent efficacy benefit.

Trial Registration: ChinaDrugTrials.org.cn identifier: CTR20221419

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Key words: Recombinant Botulinum Toxin Type A; YY001; Moderate-to-severe glabellar lines; Safety; Efficacy

Plain Language Summary

Glabellar lines, commonly known as frown lines, are the vertical wrinkles between the eyebrows caused by repeated muscle contraction and aging. This study aimed to evaluate the safety and preliminary efficacy of worldwide first recombinant BoNT/A (YY001) in humans for treatment of moderate-to-severe glabellar lines at three sites in China. Preclinical assessments comprised homology modeling, biochemical characterization, pharmacological evaluation in animal models, and multispecies toxicology studies. In this phase I multicenter double-blind, randomized, dose-ranging, vehicle- and active- controlled trial, participants (n=50) with moderate-to-severe glabellar lines were randomized 1:1:1:1:1 to receive intramuscular injection of YY001 at 10, 20 and 30U, Botox®, or vehicle. The study demonstrated a favorable safety and tolerability profile at doses of 10, 20 and 30 U in Chinese subjects with moderate-to-severe glabellar lines within 28 days post-injection, with exploratory follow-up to 168 days. Additionally, efficacy improved significantly in a dose-dependent manner.

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Introduction

In recent years, the demand for ways to prevent skin aging has increased due to the expanding global elderly population.

Clostridium botulinum toxin, a neurotoxin that inhibits the release of acetylcholine mediated by calcium, acts as a muscle paralyzer. By inhibiting muscle contraction, it can be used as a preventive or therapeutic option for wrinkles.[1, 2] Attributed to accumulated evidence demonstrating a favorable safety profile, high response rate, extended duration of effect, and high patient satisfaction,[3, 4] natural BoNT (nBoNT) products have been approved worldwide for aesthetic indications to temporarily improve facial appearance. nBoNT/A naturally occurs in a non-covalently bound progenitor toxin complex alongside accessory proteins, such as hemagglutinins (HA) and non-toxin non-hemagglutinin (NTNH).[5] These associated proteins are not involved in the paralysis induction by neurotoxin; therefore, the manufacturers of the approved nBoNT/A products, Xeomin® and Daxxify®, developed formulations based on the isolated homogeneous BoNT/A core 150 kDa protein. However, biosafety concerns exist during the production of nBoNTs because they are isolated from the pathogenic *Clostridium botulinum*.[6] Using genetic engineering technology to produce recombinant BoNT/A can significantly reduce the biosafety risks through a safer, more consistent production process. It also provides a pure therapeutic product for clinical use, eliminating the risk of administering heterogenous proteins present in most currently available BoNT therapies.

YY001, the recombinant botulinum neurotoxin type A (rBoNT/A), has been innovatively developed using genetic engineering technology by Chongqing Claruvis Pharmaceutical Co., Ltd. The rBoNT/A is a 150 kDa protein expressed in Escherichia coli (E. coli) from a synthetic gene encoding the same amino acid sequence as nBoNT/A. The manufacturing process

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incorporates a novel approach of nicking the expressed single-chain rBoNT/A into a di-chain bioactive molecule by in vitro enzyme activation, resulting in a fully bioactive rBoNT/A. Collectively, the method significantly enhanced the drug product purity, streamlined production, and eliminated the biological hazards linked to large-scale pathogen expansion. Extensive preclinical studies were conducted to characterize the recombinant BoNT/A, including comparative structural analysis with nBoNT/A, biochemical property analysis and comprehensive assessment of its pharmacological and toxicological profiles in animal models. All the preclinical study results warranted its clinical investigation. Here, we conducted the first-in-human study of YY001, reporting its safety and efficacy results for the treatment of moderate-to-severe glabellar lines.

Materials and Methods

Protein identification

Homology modelling, SDS-PAGE and Western blotting

The nBoNT/A protein sequence was obtained from the Universal Protein Knowledgebase (UniProtKB, https://www.uniprot.org/). SWISS-MODEL (http://swissmodel.expasy.org) was used to automate three-dimensional (3D) protein structure comparison of nBoNT/A and YY001. Prepared proteins were separated by SDS-PAGE using 4-20% Tris-glycine gels at 150 V for 50 minutes, followed by staining with Coomassie Brilliant Blue (Solarbio, G4540, CN) for 1 h and destaining with purified water for 3 h. For immunoblotting, resolved proteins were transferred to ethanol-activated PVDF membranes via wet transfer. The membranes were blocked overnight in 5% BSA (Solarbio, SW3051, CN), then sequentially incubated with primary antibody (Rabbit polyclonal anti-YBA antibody, 1:2000, 45 min) and secondary antibody (Goat Anti-Rabbit IgG

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H&L, 1:5000, 45 min). After extensive washing, protein signals were detected using ECL substrate (Thermo, US) and visualized with a ChemiDoc XRS+ imaging system.

Pharmacological and toxicological studies in mice and rats

Male Sprague-Dawley (SD) rats (aged 7-9 weeks, approximately 300 g, supplied by Beijing Vital River Laboratory Animal Technology Co., Ltd., China) and male CD-1 mice (aged 7-8 weeks, approximately 35 g) were used for the compound muscle action potential (CMAP) test and the digit abduction score (DAS) assay, respectively. Both male and female SD rats were used for toxicological studies. This study was approved by the Animal Ethics Committee of Chongqing Claruvis Pharmaceutical Co., Ltd. (Approval No. sz-20210629-rat). The rats and mice were housed in a controlled environment with temperatures ranging from 18 to 26°C, a relative humidity of 30-70%, and had unrestricted access to food and water. Rats (n=3 per group). In the CMAP assay, rats were anesthetized with 3-5% isoflurane (maintained at 2-3%), placed prone on a heating pad, and injected with 0.1U YY001 or Botox® in the gastrocnemius. Electrodes were placed at the fourth lumbar vertebra (stimulating), mid-gastrocnemius (recording), and tail (reference). The CMAP amplitude was recorded after adjusting stimulation current (1-20 mA) until a plateau was achieved. The CMAP ratio (%) was calculated as post-dose CMAP value divided by pre-dose CMAP value multiplied by 100%. In the DAS assay, mice (n=10 per group) had their DAS measured prior to receiving an intramuscular injection of YY001 or Botox® into the mid-belly of the right hindlimb gastrocnemius muscle. Mice were tail-suspended and lowered to elicit hindlimb extension with digits abduction. Scoring was independently performed by two blinded technicians using a standardized five-point scale to ensure an objective assessment. The score is set as follows: 4 = all five toes combined; 3 = five

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toes combined into two parts; 2 = combined into three parts; 1 = combined into four parts; 0 = not combined.[7]

Phase I clinical

Study Design

This phase I double-blind, randomized, dose-ranging, vehicle- and active- controlled study (ChinaDrugTrials.org.cn identifier: CTR20221419) was conducted across three sites in China from July 26, 2022, to February 28, 2023. The trial comprised a 28-day double-blind observed period, and a 168-day open-label follow-up period (only YY001 and Botox® groups were selected in the 168-day extension study). The overall design was shown in **Figure 1**. The study was conducted in accordance with the ethical principles of the Declaration of Helsinki, in compliance with Chinese Good Clinical Practice and the International Conference on Harmonisation-Good Clinical Practice (ICH-GCP), and was initially approved by the Biomedical Research Ethics Committee of Shanghai Ninth People's Hospital, Shanghai JiaoTong University School of Medicine (SH9H-2022-C18-3), followed by approval from the ethics committee of other study sites. All participants signed written informed consent before screening.

Study Participants

Eligible participants were Chinese male or female participants aged 18-65 years with moderate or severe glabellar lines, as assessed by the investigator using a 4-point Facial Wrinkle Scale. Participants deemed at risk for neuromuscular transmission issues following botulinum toxin injection were excluded. The main exclusion criteria are: 1) Known hypersensitivity to any component of the study products; 2) Previous cosmetic therapy in glabellar or forehead region; 3) Treatment with any botulinum toxin within 6 months pre-study or planned during trial; 4) Use of NSAIDs/anticoagulants within 1 week before treatment; 5) Use of neuromuscular-affecting

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medications within 4 weeks before screening; 6) Skin abnormalities in treatment area; 7) History suggesting neuromuscular dysfunction.

Study Treatment and Procedure

Participants were randomly assigned to YY001 10U group, YY001 20U group, YY001 30U group, Botox group, and vehicle group in a ratio of 1:1:1:1:1 using an interactive web response system with a block size of 5. The vehicle which has an identical excipient composition to YY001 but lacks the active pharmaceutical ingredient, could not be distinguished from YY001 in terms of appearance, excipients, and was packaged in the same manner. To maintain blinding, an unblinded nurse prepared the product.

Qualified participants received a single injection of YY001 (10U, 20U, and 30U; Chongqing Claruvis Pharmaceutical Co., Ltd., China), Botox® (20U, Allergan Pharmaceuticals Ireland, USA) or vehicle (Chongqing Claruvis Pharmaceutical Co., Ltd., China) following the on-label 5-site injection pattern in the glabellar muscle complex. This pattern consisted of a single 0.05-mL injection into the procerus muscle and two bilateral 0.05-mL injections into each of the corrugator muscles (total of 5 sites).

Study assessments and endpoints

The primary endpoint was the safety of the investigational drug within 28 days after injection. The safety profile, including adverse events (AEs), serious AEs (SAEs), injection site reactions, and other clinically significant examination abnormalities, was recorded and monitored throughout the trial. Efficacy was measured by response rate of improvement in glabellar lines severity at maximum frown, as assessed by the investigators or participants' self-assessment on days 7, 14, 28, 84 and 168 after injection. Glabellar lines at maximum frown were assessed using a 4-point Facial Wrinkle Scale from 0 to 3 (0 = no line, 1 = mild, 2 = moderate,

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and 3 = severe). The response was defined as the achievement of a score of 0 or 1 and at least a two-grade improvement from the baseline.

Statistical Analysis

Designed as the first human dose-ranging exploration, the study employed a sample size not derived from statistical assumptions to assess safety and explore efficacy trends and dose-response effects. All randomized participants who received at least one study product treatment were included in the safety set (SS) for safety analysis. Efficacy analysis was based on the full analysis set (FAS), defined as all randomized participants, and followed the intention-to-treat (ITT) principle. Statistical analysis was carried out using SAS version 9.4.

Results

Preclinical study

Similar protein structure of YY001 and nBoNT/A

Based on homologous modeling analysis, the 3D structure of YY001 and nBoNT/A was highly consistent (**Figure 2A**). The light chain of YY001 and nBoNT/A also exhibited a high degree of structural similarity (**Figure 2B)**. To meet molecular cloning requirements, YY001 incorporated five additional N-terminal residues in its light chain compared to nBoNT/A. However, structural analysis revealed these residues are conformationally flexible and do not perturb the toxin's overall structure or function. In addition, the disulfide bond between heavy and light chains (Cys430-Cys454) in nBoNT/A spatially overlapped with that (Cys435-Cys456) in YY001 (**Figure 2C**). Both toxins also maintained highly conserved zinc-binding pocket structures (**Figure 2D**). Furthermore, SDS-PAGE/Western blot analysis demonstrated that YY001 exhibited the same molecular weight profile as nBoNT/A: a single 150 kDa band under

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non-reducing conditions, which dissociated into 50 kDa light chain and 100 kDa heavy chain upon reduction (**Supplementary Figure 1**).

YY001 and Botox® exhibit similar pharmacodynamic profiles in animal models, as demonstrated by CMAP test and DAS assay

The CMAP ratio curves for YY001 and Botox® were parallel following intramuscular administration, and both displayed similar reversible inhibitory trends (**Figure 3A**). Within the dose range of 0.857-34.3 U/kg, a clear dose-response relationship was observed for the inhibition of DAS in the mouse model (**Figure 3B-F**). The dose-effect curves for YY001 and Botox® were parallel (**Figure 3G**). These observations indicated that YY001 and Botox® exhibited similar pharmacodynamic profiles in the animal models.

Following a single intramuscular injection of YY001 at doses of 0, 3, 10, and 30 U/kg in rats, dose-related toxicity of YY001 (≥10 U/kg) manifested as limb weakness, uncoordinated gait, and reduced grip strength, all of which were reversible by the end of the recovery phase. These behaviors were considered to be related to the amplified pharmacological effects of YY001. The potential toxicity of YY001 was evaluated following a single intramuscular injection in rats. A 12-week recovery period observation revealed no anti-YY001 antibody production and no injection site irritation. The no-observed-adverse-effect level (NOAEL) was determined to be 30 U/kg.

Phase I clinical

Participant disposition and characteristics

A total of 50 participants were randomly assigned to YY001 10U group (n=10), YY001 20U group (n=10), YY001 30U group (n=10), Botox® group (n=10) and vehicle group (n=10), and all completed a 28-day follow-up (**Figure 4**). Baseline demographic and characteristics were

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shown in **Table 1**. Of the 50 participants in the FAS population, 64% were female and 94% were of Han ethnicity, with a mean ± SD age of 38.4 ± 10.59 years. All treatment groups demonstrated similar demographic characteristics, although minor variations existed in baseline glabellar line severity at maximum frown due to small group sizes.

Safety

Safety outcomes through Day 28 post-injection for all 50 participants are summarized in **Table 2**. All adverse events (AEs) were mild or moderate in severity according to CTCAE criteria, and the incidence and severity of AEs did not show any dose-response effect for any category of AEs related to YY001. The incidence of AEs of all participants in three YY001 groups was 26.7% (YY001 10U group, 30.0%; YY001 20U group, 30%; YY001 30U group, 20%), which was numerically lower than that in the active control group (50%). No AEs occurred in the vehicle group. There were no SAEs, injection site reactions, CTCAE grade ≥3 AEs, acute AEs, or AEs leading to study withdrawal that occurred throughout the study. The incidence of clinically significant abnormal laboratory changes was 10.0%, 20%, 10% and 20% in YY001 10U group, YY001 20U group, YY001 30U group and Botox® group, respectively. None of these laboratory value abnormalities was related to the investigational drug. Two participants of YY001 20U group experienced facial pain and facial swelling, respectively. Additionally, one participant of YY001 30U group had an episode of elevated blood pressure. All events were resolved without sequelae.

Furthermore, the safety assessments in YY001 groups and Botox® group were also conducted within 28-168 days after injection. No SAEs, drug-related AEs, or AEs leading to study withdrawal occurred within 28-168 days after injection.

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Efficacy

The response rate of improvement in glabellar lines at maximum frown on day 28 by investigator's live assessment was 60% in YY001 10U group, 80.0% in YY001 20U group, 90.0% in YY001 30U group, 70.0% in Botox® group, and 0.0% in vehicle group. According to participant's self-assessment, 40.0% of participants in YY001 10U group, 70.0% of participants in YY001 20U group, 70.0% of participants in YY001 30U group, 40.0% of participants in Botox® group, and 0.0% (0.0%, 30.8%) of participants in vehicle group responded at maximum frown (**Table 3 and Figure 5**). At Day 168 after injection, 60.0%, 22.2%, and 20.0% of subjects in the YY001 10 U, 20 U, and 30 U dose groups, respectively, maintained dynamic glabellar line severity at none or mild by investigator's live assessment (**Figure 5**).

Discussion

YY001, as the world's first recombinant BoNT/A in clinical trial, has a protein structure and physicochemical properties that are highly consistent with those of nBoNT/A. The preclinical data have shown that YY001 possesses comparable bioactivity and toxicity characteristics to nBoNT/A. Furthermore, this first-in-human study demonstrated that doses of YY001 from 10U to 30U were safe and well-tolerated in the treatment of moderate-to-severe glabellar lines, with no significant dose-dependent effects on safety outcomes. The response rate of improvement in glabellar lines at maximum frown to YY001 was observed to increase in a dose-dependent manner. Overall, this study provided promising evidence that YY001 was safe and effective in treating moderate-to-severe glabellar lines in Chinese participants.

YY001, a recombinant BoNT/A, replicated the full amino acid sequence of nBoNT/A and exhibited a highly consistent 3D structure. The key disulfide bond position between the light chain and the heavy chain is highly comparable between YY001 and nBoNT/A, which provided

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the basis of the toxin bioactivity.[8] The highly conserved zinc-binding pocket between nBoNT/A and YY001 provided the essential feature for the comparable enzymatic activity since the proteolytic activity of the BoNT/A light chain is zinc-dependent, with Zn^{2+} serving a catalytic role.[9, 10] This high degree of structural conservation served as the foundation for the favorable biological activity of YY001 observed in preclinical and clinical studies. In addition, the CMAP test and DAS assessment acted as effective pharmacodynamic (PD) indicators to evaluate the pharmacological activity of BoNT-based products.[11] The preclinical study results also demonstrated comparable efficacy and duration of action between YY001 and Botox®, suggesting therapeutic equivalence between the two BoNT/A preparations.

In this study, the participants in YY001 group did not experience the drug-related adverse reactions that often occurred in other BoNT/A products at the same dose level, such as headache.[12-14] Only one case of drug-related eyelid ptosis occurred in Botox® group—the most concerning adverse event resulting from unintended BoNT diffusion.[15, 16] The favorable safety profile of YY001 demonstrated in this clinical study indicated that its high purity compared to most of the nBoNT/A products on the market may be a key contributing factor. The preliminary efficacy analysis indicated that both YY001 and Botox® began to show effects within the first week following administration, with peak effects observed on Day 28. This pharmacodynamic profile is consistent with that of nBoNT/A.[17] Furthermore, the investigator-assessed response rate for YY001 at a dose of 20U was 80%, which was comparable to published nBoNT/A data: BOTOX® 20U (76.2%[18]), Daxxify™ 20U (73.5%[18]) and Dysport® 50U (63.6%[14]). Numerically, the low dose of YY001 (10U) demonstrated lower response rates than higher-dose cohorts (20U and 30U) and Botox® 20U, suggesting a suboptimal efficacy. However, the dose-dependent relationship between the 20 U and 30 U groups, evident at the peak therapeutic effect (Day 28),

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appeared attenuated by the 6-month follow-up (Day 168), a finding that may be attributable to the limited sample size. Twenty percent of participants maintained glabellar lines severity at none or mild on day 168 after injection with 20 or 30 U of YY001. This result offers a new treatment alternative for patients seeking long-term, optimal outcomes for glabellar lines. Based on the balance of risk and benefit,[19, 20] as well as the common dosage of the active control drug, YY001 at 20U can be an appropriate dosage for the treatment of moderate to severe glabellar lines.

Our study implemented several key measures to minimize the likelihood of bias. To preclude selection bias, participants were randomly allocated to one of the five treatment groups using an interactive web response system with a block size of 5, ensuring balanced and unpredictable allocation. To reduce performance and detection bias, rigorous blinding was applied: YY001 and the vehicle were identical in appearance, excipient formulation, and packaging. An unblinded nurse prepared the investigational products, ensuring that participants, investigators, and outcome assessors remained blinded to treatment assignments throughout the study. This design effectively minimized the potential for bias in outcome reporting and in the assessment of efficacy and safety endpoints.

Despite the robust design, several potential limitations should be considered. First, although blinding was rigorously maintained for the intervention, the response rate of improvement based on the participant's self-assessment may still be prone to subjective bias. Second, in the absence of statistical significance testing, the numerical differences between groups observed in the report should be interpreted as descriptive trends rather than confirmatory efficacy conclusions. The significance and stability of these trends require validation through future rigorously designed confirmatory trials with adequate statistical power. In addition, definitive conclusions

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regarding the long-term safety of YY001 will require verification in larger, prospectively designed confirmatory trials with extended follow-up durations. Finally, as is the case with any clinical trial, the highly controlled environment and selected patient population may not fully reflect real-world practice, where treatment outcomes can be influenced by a wider range of factors.

Due to the small number of participants in this study, no significant differences in efficacy were observed between the toxin group and the vehicle group. Nevertheless, this study firstly provides the initial clinical evidence regarding the application of the recombinant BoNT/A for the treatment of moderate-to-severe glabellar lines, supporting the need for further evaluation. Currently, both phase II and phase Ⅲ clinical trials have been completed (ChiCTR2400088794; NCT06499688), and an additional open-label repeated treatment phase Ⅲ study (NCT06583486) is ongoing. All the studies will improve the understanding of the safety, efficacy, and immunogenicity of YY001 in patients with moderate-to-severe glabellar lines.

Conclusions

The first recombinant BoNT/A tested in humans, which is structurally and functionally similar to natural BoNT/A, was deemed safe, well-tolerated, and effective for treating moderate-to-severe glabellar lines at doses of 10U, 20U, and 30U. Notably, there was no apparent impact of dose on safety profile, whereas it showed a dose-dependent pattern in clinical efficacy.

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Figure Legends

Figure 1 Study design

Figure 2 Three-dimensional structure of YY001and botulinum toxin type A (BoNT/A) based on homologous simulation. (A) Overall structure (B) Light chain structure (C) Disulfide bond space region (D) Zinc-binding site structure YY001 was represented by orange, and nBoNT/A was represented by blue.

Figure 3 Inhibition of compound muscle action potential (CMAP), and digit abduction score (DAS) of muscle tissue at injection site after YY001 and Botox® intramuscular injection. (A) CAMP ratio (B-F) DAS under different dose injection (G) Half maximal inhibitory concentration (IC_{50}) of DAS at 24 h post-dosing.

Figure 4 Phase I Study flow

Figure 5 Response rate of improvement in glabellar lines at maximum frown by investigator's live assessment (A) and participant's self-assessment (B) on Day 7, 14, 28, 84 and 168 after injection

Supplementary Figure Legends

Supplementary Figure 1 The molecular weights of YY001 under reducing and non-reducing conditions as analyzed by SDS-PAGE/Western blot. M, marker; 1, YY001 under non-reducing conditions (SDS-PAGE); 2, YY001 under reducing conditions (SDS-PAGE); 3, YY001 under non-reducing conditions (WB); 4, YY001 under reducing conditions (WB).

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Table 1 Demographic and baseline characteristics

Characteristic	YY001 10U (n=10)	YY001 20U (n=10)	YY001 30U (n=10)	Botox (n=10)	Vehicle (n=10)	Total (n=50)
Age, year	35.8±10.34	35.4±11.68	41.0±8.91	42.7±10.53	38.2±11.46	38.6±10.59
Sex, n (%)						
Male	5 (50)	5 (50)	4 (40)	3 (30)	1 (10)	18 (36)
Female	5 (50)	5 (50)	6 (60)	7 (70)	9 (90)	32 (64)
BMI (kg/m^2)	22.88±2.885	22.27±2.741	26.00±5.732	22.30±2.769	21.39±3.742	22.97±3.937
Race, n (%)						
Han ethnicity	9 (90)	8 (80)	10 (100)	10 (100)	10 (100)	47 (94)
Other	1 (10)	2 (20)	0 (0)	0 (0)	0 (0)	3 (6)
Past history, n (%)						
Prior drug history	1 (10)	1 (10)	0 (0)	2 (20)	1 (10)	5 (10)
Prior non-drug treatment	3 (30)	2 (20)	2 (20)	3 (30)	0 (0)	10 (20)
Investigator-rated GLS, n (%)						
Moderate	8 (80)	3 (30)	5 (50)	3 (30)	7 (70)	26 (52)
Severe	2 (20)	7 (70)	5 (50)	7 (70)	3 (30)	24 (48)
Participant-rated GLS, n (%)						
Moderate	6 (60)	2 (20)	2 (20)	4 (40)	9 (90)	23 (46)
Severe	4 (40)	8 (80)	8 (80)	6 (60)	1 (10)	27 (54)

Abbreviation: BMI, body mass index; GLS, glabellar lines severity.

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Table 2 Summary of AEs within 28 days after injection

AEs, n (%)	YY001 10U (n=10)	YY001 20U (n=10)	YY001 30U (n=10)	YY001 total (n=30)	Botox (n=10)	Vehicle (n=10)
Any AEs	3 (30)	3 (30)	2 (20)	8 (26.7)	5 (50)	0 (0)
CTCAE grade 1	1 (10)	2 (20)	1 (10)	4 (13.3)	4 (40)	0 (0)
CTCAE grade 2	2 (20)	1 (10)	1 (10)	4 (13.3)	1 (10)	0 (0)
CTCAE grade≥3	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Drug-related AEs	0 (0)	0 (0)	0 (0)	0 (0)	1 (10)	0 (0)
Eyelid ptosis	0 (0)	0 (0)	0 (0)	0 (0)	1 (10)	0 (0)
Acute AEs*	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
SAEs	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Injection site AEs	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
AEs leading to study withdrawal	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)

*Occurred within 30 min of administration

Abbreviation: AEs, adverse events.

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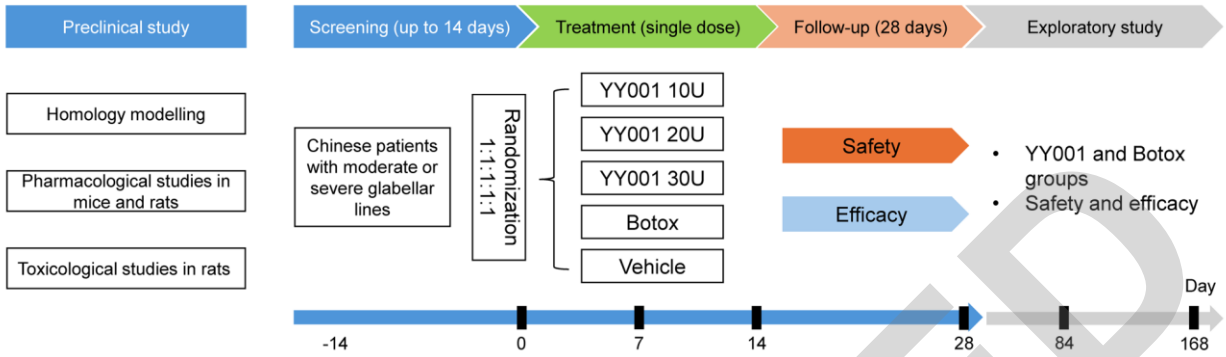
Table 3 Response rate of improvement in glabellar lines at maximum frown by investigator's live assessment and participant's self-assessment on day 28 after injection

	YY001 10U (n=10)	YY001 20U (n=10)	YY001 30U (n=10)	Botox (n=10)	Vehicle (n=10)
Based on investigator's live assessment					
28 days after injection					
Score of 0 or 1	8 (80)	9 (90)	10 (100)	8 (80)	0 (0)
Two-grade improvement	6 (60)	8 (80)	9 (90)	7 (70)	0 (0)
Response rate (%, 95% CI)	60 (26.2, 87.8)	80 (44.4, 97.5)	90 (55.5, 99.7)	70 (34.8, 93.3)	0 (0, 30.8)
Based on participant's self-assessment					
28 days after injection					
Score of 0 or 1	8 (80)	9 (90)	8 (80)	6 (60)	0 (0)
Two-grade improvement	4 (40)	7 (70)	7 (70)	4 (40)	0 (0)
Response rate (%, 95% CI)	40 (12.2, 73.8)	70 (34.8, 93.3)	70 (34.8, 93.3)	40 (12.2, 73.8)	0 (0, 30.8)

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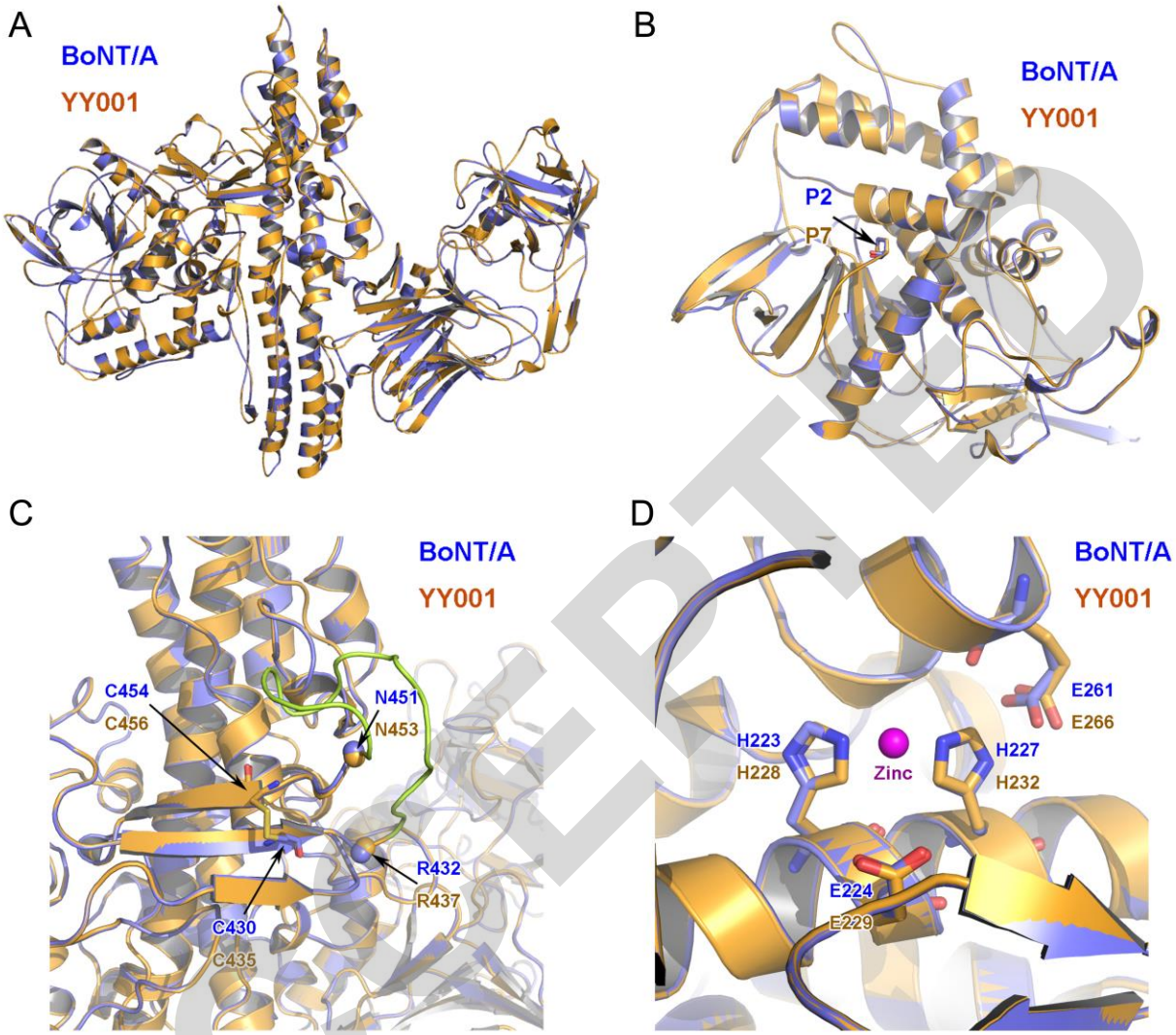
Figure 1



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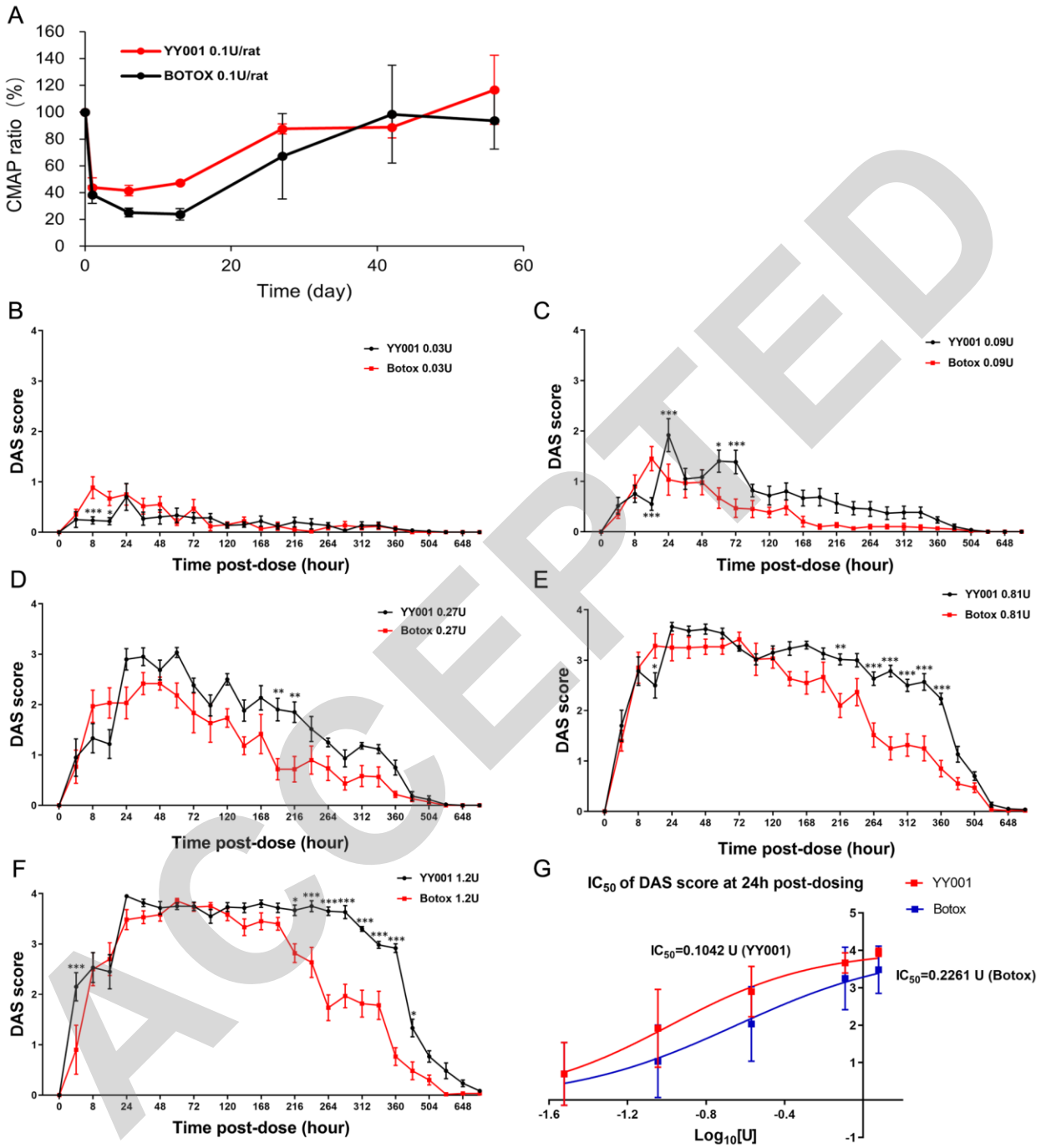
Figure 2



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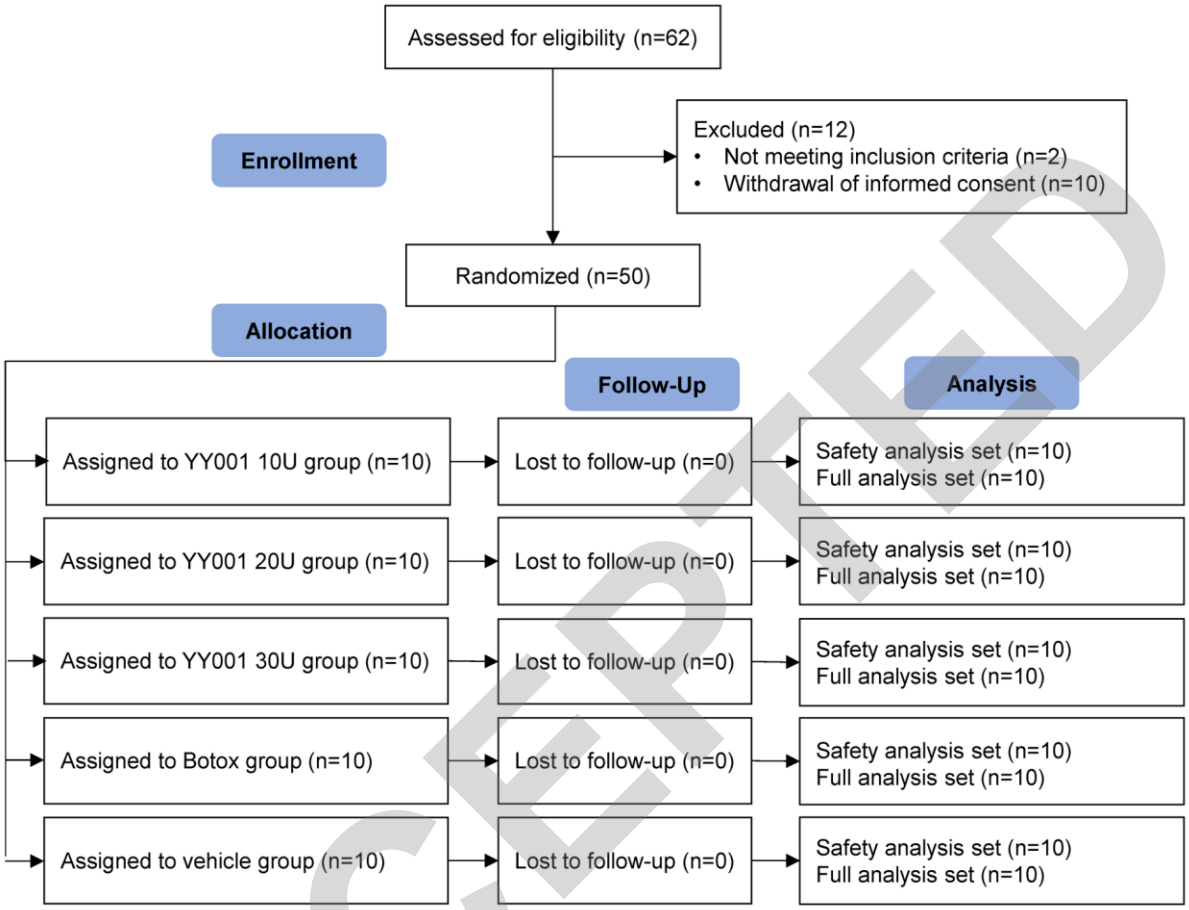
Figure 3



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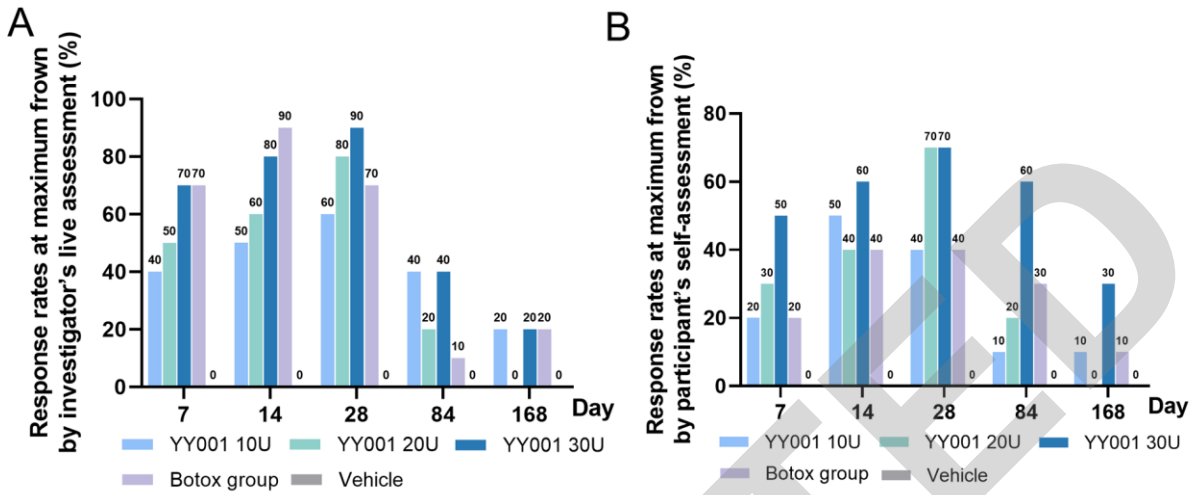
Figure 4



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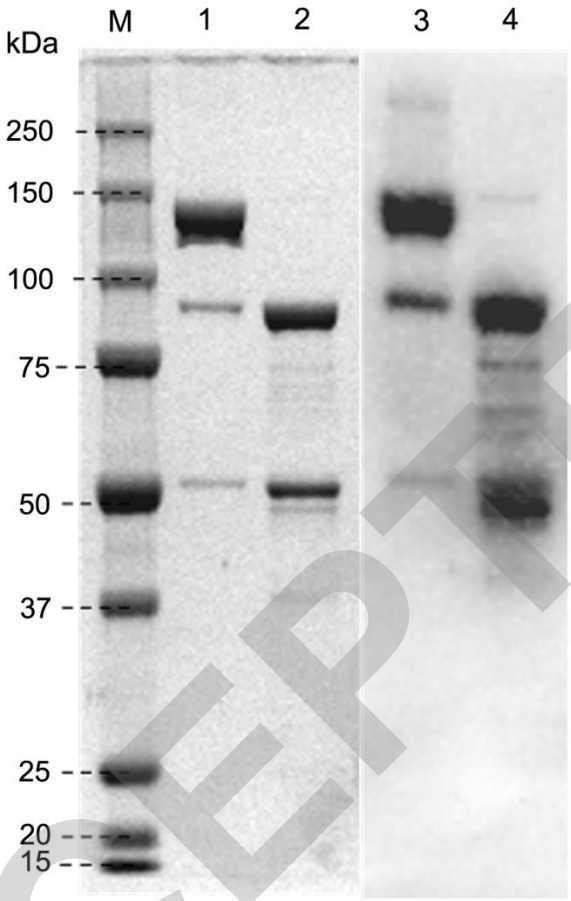
Figure 5



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SDC Figure 1



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